

NEUROMORPHIC COMPUTING

Thank you for publishing the 'Neuromorphic Computing: It's Not Sci-Fi Anymore' article in February issue. It is another revolutionary idea which might overturn our world!

Chi Tung Mak

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AI CAMERA FOR TRACKING

The 'AI Camera For Tracking Suspicious Activities' DIY article published in March issue is really a very good one!

Bhuvan Vyashu

EFY. Thanks to the readers for their feedback, which keeps us going!

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TOUCHLESS DOORBELL

This is regarding the 'Touchless Doorbell For The Post-Covid World' DIY article published in September 2021 issue. I am interested in building this project. It will be very helpful if you could help me by sending its PCB and component layouts in .pdf format.

Shristi

EFY. We are arranging it for you. In future, please send such requests through support@efy.in

HEAT-SINK

Why heat-sink details, including their dimensions, are not included in DIY articles? And how to select the heat-sinks for the ICs?

Anirvan Kule

EFY. EFY has covered several articles on heat-sinks earlier. However, some general information is being given here.

The size of a heat-sink depends not

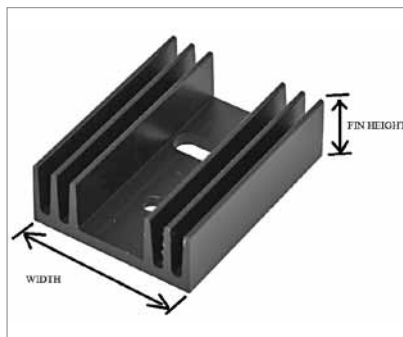


Fig. 1: A typical heat-sink for TO-220 package

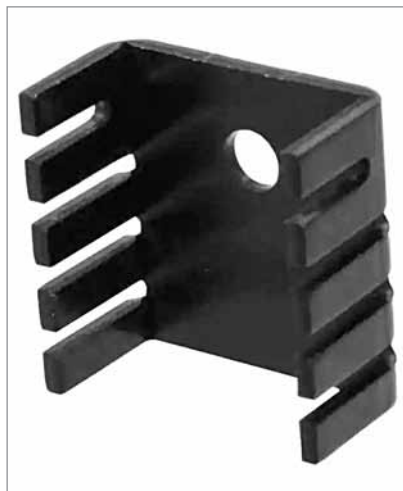


Fig. 2: A square shaped heat-sink for TO-220 package

only on a particular device but also on its applications. Though heat-sink dimensions are not included in most datasheets, important parameters such as power dissipation, thermal resistance, junction temperature, and load current of a particular device or IC are mentioned in the datasheets. These parameters help decide the approximate dimensions of the heat-sinks.

The junction temperature in a power-handling device rises in proportion to the current consumption. The thermal resistance of the device is important because most electrical components generate heat and need to be cooled. The main goal of the heat-

sink is to maintain the temperature below the specified maximum operating temperature of the device.

Selecting a heat-sink becomes simpler when you have the datasheet of the device or regulator IC. You can choose it as per the device's package, requirements, and applications. There are many readymade heat-sinks available in the market. The most common types of heat-sinks for TO-220 package regulators have natural convection cooling with or without fan.

Regulator ICs like LM317, LM78xx, LM79xx, and L200C are available in TO-220 packages. Fig.1 shows width and fin height of natural convection heat-sink for TO-220 package IC. It is an extrusion TO-220 package type heat-sink with the dimension of 35mm × 29mm × 12mm (length, width, fin height), suitable for use with a device having thermal resistance up to 15.2°C per watt.

Aluminium heat-sinks having large width rather than greater fin height are preferred. Also, if the heat-sink surface is anodised or painted, it increases heat radiation.

A commonly used heat-sink for 2.5W power dissipation at 60°C and thermal resistance up to 24°C per watt is shown in Fig. 2. It is a black anodised aluminium type heat-sink with dimensions of 19.05mm × 19.05mm × 9.65mm height. For more information on heat-sinks, you may visit following three websites:

1. <https://www.electronicspecifier.com/news/blog/how-to-specify-the-optimal-heat-sink-for-your-design>
2. https://www.nutsvolts.com/magazine/article/how_to_intro_to_heat-sink_selection_and_installation
3. <https://www.electronics-cooling.com/1995/06/how-to-select-a-heat-sink>

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